

No file came through — looks like the `.md` didn't attach. Drop it in the chat and I'll do a full gap analysis against the SENTINEL PRD, flag what's redundant, what's still missing, and what new tools have changed the game since we mapped the stack.

SENTINEL_PRD.
md

MD

this files what other new features related

Read the uploaded PRD to see what tools/features have been added >

Good — I can see the full v2.0 PRD. This is the same document we generated. You haven't added new tools yet — the file is identical to what we produced. So you're asking: **what's still missing from this stack that should be added?**

Let me do a proper gap analysis across 6 dimensions a senior OSINT pro would immediately notice are absent.

GAP 1 — COMPLETELY MISSING Face, image & video forensics



PimEyes / FaceCheck.id paid API

Reverse face search across billions of public web images. Find who is in a photo across news sites, social media, videos.

Missing: your PRD has zero face/person identification. Propaganda often uses unidentified people — this names them.

ffmpeg + OpenCV free

Extract keyframes from videos, detect scene changes, run reverse image search on each frame. Unmask recycled footage.

Missing: yt-dlp downloads video but you never analyse frames. Old footage repackaged as new is the #1 visual misinfo technique.

Video perceptual hash free

Detect same video re-uploaded with trimmed edges, filter strips, watermarks removed. videohash or frame-by-frame pHash.

Your imagehash only covers static images. Videos are re-uploaded thousands of times with minor edits to evade detection.

InVID / WeVerify API free

EU-funded video verification tool. Keyframe extraction, reverse search, metadata analysis, magnifier. Direct API available.

Industry standard for journalists. Every major fact-checking org uses this. Should be in your pipeline.

GAP 2 — COMPLETELY MISSING Dark web & paste site monitoring

Paste site monitor free

Pastebin, Ghostbin, Rentry, HasBin. Coordinated ops often pre-stage content here before social push. Raw paste APIs available.

Missing entirely. Paste sites are the pre-launch staging ground for many disinformation ops.

Ahmia / OnionSearch free

Search Tor-indexed public content via Ahmia.fi clearnet API. Finds dark web forums discussing narratives without Tor access.

State actors and extremist groups coordinate on dark web. Ahmia indexes public onion content via clearnet — no Tor needed.

HIBP / breach data API key

Email addresses found in domains/WHOIS can be cross-referenced against breach databases to find linked accounts.

WHOIS registrant emails traced through breach data is one of the most powerful attribution chains in real OSINT work.

Matrix / Element monitor free

Matrix protocol public rooms. After Telegram bans, ops migrate to Matrix. matrix-nio Python client, no auth for public rooms.

Your platform covers Telegram but not its successor. After every major Telegram ban wave, activity shifts to Matrix.

GAP 3 — COMPLETELY MISSING DNS & network attribution (deeper)

SecurityTrails free tier

Passive DNS history, subdomain enumeration, WHOIS history, IP neighbours. Far more comprehensive than crt.sh alone.

Your stack has crt.sh but misses passive DNS history. SecurityTrails shows what IPs a domain resolved to over its entire lifetime.

RIPEstat / BGP.he.net free

Full BGP routing history, ASN ownership chain, IP prefix announcements. Who actually owns the network, not just who hosts on it.

Shodan shows the server. RIPEstat shows who controls the network routing. Critical for state-actor attribution.

GreyNoise API free tier

Classifies IPs: benign scanner, malicious, unknown. Find if IPs in your infrastructure clusters are known threat actors.

Missing from your infra layer. Tells you whether a hosting IP is associated with known malicious infrastructure.

WHOISXIVL API

free tier

Bulk WHOIS + historical WHOIS + reverse WHOIS (find all domains registered by same email/phone). 10,000 free queries/month.

Reverse WHOIS is the killer feature — one registrant email → all their domains. Your current python-whois can't do this.

GAP 4 — WEAK IN PRD AI / ML upgrades to existing modules

OpenAI Whisper

free / local

Best-in-class multilingual audio transcription. Transcribe Telegram voice messages, YouTube videos, TikTok audio in Urdu/Hindi/Bengali locally.

Your stack uses yt-dlp subtitles only, misses auto-transcribed content. Whisper natively

AI image detection

2025 critical

Hive Moderation API or local models (CNNDetection, UniversalFakeDetect) flag AI-generated / deepfake images. Free tiers available.

Not in your PRD yet! AI-generated

ClaimBuster / ClaimReview

free API

ClaimBuster scores claim check-worthiness. ClaimReview schema links to existing fact-checks from Snopes, AFP, AltNews, etc. Auto-attach to results.

How do I run EXIF GPS extraction and geolocation verification on images in Python for OSINT investigations?

Perspective API

free



Google's toxicity scoring API. Scores: toxicity, severe toxicity, insult, threat, identity attack. Adds manipulation intensity metric per result.

Your sentiment analysis is basic. Perspective adds a manipulation intensity dimension that maps directly to DISARM T0006.

GAP 5 — WEAK IN PRD Live monitoring & alerting (you have no watchdog)

Watchdog daemon

build it

Background Celery beat scheduler re-runs tracked narratives every N minutes. Alerts when velocity crosses threshold. Push via Slack/email/Telegram bot.

Your PRD is all reactive. A pro tool monitors continuously. You should know about a spike before the analyst opens the tool.

feedparser + changedetection

free

feedparser polls 100+ RSS feeds. changedetection.io monitors non-RSS pages for content changes. Both feed into Celery pipeline.

Your news source packs are one-shot searches. You need continuous RSS ingestion so new articles auto-enter the evidence pool.

python-telegram-bot alerts

free

Send velocity spike alerts, new evidence locked, infrastructure cluster found — as Telegram messages to analyst. Two-way: reply to trigger actions.

Analysts aren't staring at SENTINEL all day. Push alerts when something important happens.

newspaper4k / trafilatura

free

trafilatura (better than newspaper3k) extracts clean article text, author, date, language from any URL. Massive upgrade over BeautifulSoup scraping.

Your Playwright scraper gets raw HTML. trafilatura gives you clean structured article data in one line. Use it for all news sources.

GAP 6 — PROFESSIONAL GRADE UPGRADES What separates amateur from pro OSINT tools

Maltego TRX export

[pro signal](#)

Export your NetworkX graph as a Maltego-compatible .mtgx file. Every serious OSINT team uses Maltego. Native export makes SENTINEL interoperable.

No serious analyst will trust a tool that can't export to Maltego. One export function and you're in their workflow.

stix2 Python library

[free](#)

Create proper STIX 2.1 objects: Campaign, ThreatActor, Infrastructure, Relationship. Export to TAXII server or share as STIX bundle JSON.

PRD mentions STIX but no library. stix2 is the official Python SDK. Needed for government/NGO sharing of intelligence findings.

MISP integration

[free](#)

MISP is the standard platform for sharing threat intelligence. pymisp lets you push SENTINEL findings into MISP instances used by CERTs and governments.

Governments and NGOs running MISP can receive your findings automatically. Makes SENTINEL a feeder into existing intel ecosystems.

Meilisearch (replace ES)

[free / local](#)

Meilisearch is Elasticsearch but runs in 50MB RAM with zero config. Full-text + faceted search across all evidence. Perfect for local SENTINEL installs.

Elasticsearch needs 2GB RAM minimum — kills local installs. Meilisearch does everything you need at 1/40th the resource cost.

SpiderFoot modules

[open source](#)

SpiderFoot has 200+ OSINT modules. Run it as a subprocess or import modules directly. Auto-correlates IPs, domains, emails, names, usernames.

Why rebuild OSINT modules from scratch when SpiderFoot has 200 already? Import its modules into SENTINEL instead of reinventing them.

Confidence scoring engine

[build it](#)

Each finding gets a 0–100 confidence score: semantic match %, source credibility, corroboration count, infrastructure signals, bot score. Weighted composite.

PRD shows individual signals but no unified confidence score. Analysts need one number per finding to prioritise investigation time.

New libraries to add to requirements.txt